

* List comprehension:

⇒ A concise way to create a new list by processing each element of an iterable.

Syntax

[expression for item in iterable if condition]

Ex: 1, 2, 3, 4, 5

Squares = [i*i for i in range(1, 6, 1)]

Squares

o/p [1, 4, 9, 16, 25]

Ex 1: Even Numbers

even = [i for i in range(1, 11, 1) if i%2 == 0]

o/p [2, 4, 6, 8, 10]

* Tuple

⇒ It is also multivalued variable which stores heterogeneous data and ordered, immutable, allow duplicates.

Tuple defined by : c) or tuple c)

Ex: T1 = ("Kritan", 21, True, 5+2j, 12.76, 45, 21)

o/p ⁺
'Kitan', 21, True, (s+2j), 12.76, 45, 21)

we printing sequence

for i in +1:
 print(i)

o/p

Kitan	True	12.76	21
21	(s+2j)	45	

note we can't do modify because tuple is a immutable.

- we can purposefully modifying tuple we must convert into the tuple in to list after we can convert list into tuple.

⇒ tuple to list

L1 = list (+1)

['Kitan', 21, True, (s+2j), 12.76, 45, 21]

we add in the True place to False

L1[2] = False

L1

['Kitan', 21, False, (s+2j), 12.76, 45, 21]

we convert list into tuple

t1 = tuple(L1)

t1

['kiran', 21, False, (5+2j), 12.76, 45, 21]

* Set

⇒ It is also multivalued variable heterogeneous datatype and unordered, unindexed, mutable and contains only unique items (no duplicates).

set defined by : {} and set()

Ex: s1 = {'c', 12, 23, 56, True, False, True}

s1

o/p {12, 23, 56, 'c', False, True}

Note

we can't extract values through indexed number because the set is a unordered or unindexed.

Ex1: s1 = {1, 2, 3, 4, 5}

s2 = {4, 5, 6, 7, 8}

s1.union(s2)

o/p {1, 2, 3, 4, 5, 6, 7, 8}

Ex2: s1.intersection(s2)

o/p {4, 5}

Ex3: s1.add(10)

s1

o/p {1, 2, 3, 4, 5, 10}

Ex4: s1.remove(10)

s1

o/p {1, 2, 3, 4, 5}

Ex5: we can delete the set1 or set2

s1.clear()

s1

o/p set()

Ex6: s1.difference(s2)

o/p {1, 2, 3}

Ex7: s2.difference(s1)

o/p {6, 7, 8}